

Factory Acceptance Test

Factory Acceptance Test Procedure

PRODUCT NAME	HiNAS Control Standard	SERIAL NUMBER	MS25-0026
YARD	HHI	HULL NO.	3447
CLASS	DNV	DATE	2026-05-21
INSPECTOR	LEE Jong Hak		

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Version	Description	Author/EDITED	Revised/Approved	Date
1.0	First Draft	Dongheun Kang, Jinhwan Park	Sangdeok Choi	2025-07-02
1.1	Minor Update	Dongheun Kang, Jinhwan Park	Sangdeok Choi	2025-07-03
1.2	Comment Update	Dongheun Kang	Sangdeok Choi	2025-07-10

Note: In this document, ‘HiNAS Control’ is used to denote both HiNAS Control and HiNAS Control Standard.



Factory Acceptance Test

1 Overview

1.1 Purpose

This document provides a comprehensive description of the Factory Acceptance Test (FAT) for HiNAS Control Standard. The following items are verified through this document:

1. Verify that the configuration and quantity of the HiNAS Control Standard hardware are the same as those shown in the drawings (Appendix A) submitted with the MED-B certificate.
2. Verify that the Serial Numbers of the hardware listed in the DoC document match those of the hardware actually installed onboard.
3. Verify that the software version recorded in the SW install package used for installation matches the version in the actual product.
4. Verify that the data from all navigation equipment connected to the overall system are received correctly and that the software functions operate properly.
5. Verify that the connected hardware performs its functions correctly.

1.2 Test Environment

Unless otherwise specified, the following test conditions shall be applied uniformly during the Factory Acceptance Test (FAT):

- Temperature: 25 °C ± 10 °C
- Atmospheric Pressure: Normal ambient pressure
- Humidity: 20% to 80%

1.3 Test Site

The test shall be conducted on board

1.4 Applicable Document

1.4.1 EC-TYPE EXAMINATION CERTIFICATE (MODULE B)

MEDB0000BWN (Revision no. 1)

1.4.2 Drawing

Approval Drawing of the vessel subject to FAT. (Appendix A.)

1.4.3 Inspection Report

Inspection Report for the item subject to inspection

1.4.4 SW Installation Report

SW Installation Report for the item subject to software installation



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2 Information about the test setup

2.1 Components List (MEDB0000BWN)

The components list specified in the EC-Type Examination Certificate is as follows.

No.	Type	Type Designation	Maker	Note/description/limitation
1	Server	Optiplex XE4 Tower with damping kit TVK-XE4-TV or TVK-XE4-TH	Tetradyne	At least two per installation.
2	Client	Optiplex XE4 Tower with damping kit TVK-XE4-TV or TVK-XE4-TH	Tetradyne	Optional. In server/client config, two per installation
3	Monitor	TIMX-W270 or TIMX-W320 or TIMX-UW584	Tetradyne	System may be installed with two monitors in same installation.
4	Keyboard with Trackball	RKTE85B2448- W-MC1	NSI	To be installed with each delivery
5	Serial to Ethernet	Nport-5450i	Moxa	Minimum one per installation
6	Network switch	EDS-G508E	Moxa	Minimum one per installation
7	Firewall	FGR-60F	Fortinet	One per installation
8	Selector switch	MPH 731	Yokogawa	One per installation
9	Dimmer	MKD002	Yokogawa	One per installation
10	UPS	9SX1000IM or 9SX3000IM or 9PX1500iRTM	Eaton	Minimum one UPS per installation.
11	Camera unit, incl. EO cameras and IR cameras	HiNAS Navigation	Avikus	Optional
12	Junction box incl. Single Board Computer			

2.2 Components' Software Version

No.	Type	Type Designation	Software Version
1	Heading Control System	PR-9000	HCCPU VER.2.00
2	Remote Control System	M-800-VII	2.0.1
3	ECDIS	NDC-1590A	01.60
4	HiNAS Control Standard		V2.0.4
5	Operating System	ubuntu	22.04

2.3 Test Preparation

2.3.1 Verification

Verify whether there are any non-compliant components by comparing the drawings of the FAT target vessel with the EC-Type Examination Certificate.

2.3.2 Installation

All components shall be fully installed on the test bed prior to testing.



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2.3.3 Wiring

The test items shall be wired according to the wiring diagram of the FAT target vessel.

The selector switch and dimmer box are excluded from the wiring, as there are no external interface devices.

2.3.4 Power Input

After verifying that there are no wiring errors, system power shall be applied.

2.3.5 Software Installation

The SW is installed by 'Manual: HiNAS Control Standard Installation Manual' on MED-B cert.



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3 Factory Acceptance Test

3.1 Visual Inspection Test

3.1.1 Visual Inspection

No.	Test	Criteria	Result	Judgment	
				Pass	Fail
1	Visually inspect the prepared test items for any visible damage or deformation	No damage or deformation		v	

3.1.2 Verification of conformity

Verify that the FAT items conform to the Type Designation specified in the EC-Type Examination Certificate

※ For items not applicable, indicate "N/A" in the Result column and skip the test.

No.	Type	Type Designation	Qty	Criteria	Result	Judgment	
						Pass	Fail
1	Server	Optiplex XE4 Tower with damping kit TVK-XE4-TV or TVK-XE4-TH	2	Type Designation conformity	Type Designation : Optiplex XE4 Tower with damping kit TVK-XE4-TH SER. NO. : G1RXL54 60RXL54	v	
2	Client	Optiplex XE4 Tower with damping kit TVK-XE4-TV or TVK-XE4-TH	1	Type Designation conformity	Type Designation : Optiplex XE4 Tower with damping kit TVK-XE4-TV SER. NO. : 2HDXL54	v	
3	Monitor	TIMX-UW584	1	Type Designation conformity	Type Designatio : TIMX-UW584(58.4") SER. NO. : TM250814-024	v	
4	Keyboard with Trackball	RKTE85B2448-W-MC1	2	Type Designation conformity	Type Designation : RKTE85B2448-W-MC1 SER. NO. : 2025.39630.040	v	



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No.	Type	Type Designation	Qty	Criteria	Result	Judgment	
						Pass	Fail
5	Serial to Ethernet	Nport-5450i	2	Type Designation conformity	Type Designation : Nport5450i SER. NO. : 6347 6377	v	
6	Network switch	EDS-G508E	2	Type Designation conformity	Type Designation : EDS-G508E SER. NO. : TBEDD1028848 TBECD1089468	v	
7	Firewall	FGR-60F	1	Type Designation conformity	Type Designation : FGR-60F SER. NO. : LNSBEJEBFC	v	
8	Selector switch	MPH 731	1	Type Designation conformity	Type Designation : MPH 731 SER. NO. : 0962	v	
9	Selector Switch Dimmer	MKD002	1	Type Designation conformity	Type Designation : MKD002 SER. NO. : 11321	v	
10	UPS	9PX1500iRTM	1	Type Designation conformity	Type Designation : 9PX1500iRTM SER. NO. : GA41T06007	v	

No.	Type	Type Designation	Qty	Criteria	Result	Judgment	
						Pass	Fail
11	Camera unit, incl. EO cameras and IR cameras	HiNAS Navigation	1	Type Designation conformity	Type Designation: HiNAS Navigation SER. NO. : Camera unit : SC0004-NCH01-25108	v	
12	Junction box, incl. Single Board Computer		1	Type Designation conformity	Type Designation: HiNAS Navigation SER. NO. Junction box : SC0005-NJB01-25108 Single board computer : NAS-HUCX0620065 EMC -filter : SZ0014-EMC01-046 SMPS : 2025-030	v	



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Verify that the types and quantities of equipment installed on board the vessel match those specified in the Declaration of Conformity and the submitted drawings.

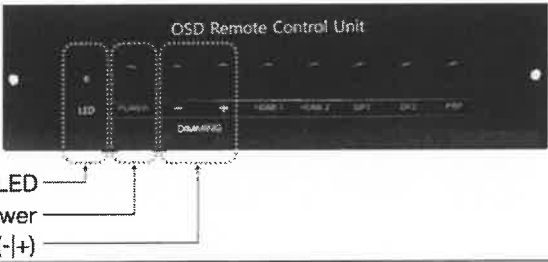
No.	Test	Criteria	Result	Judgment	
				Pass	Fail
1	Verify that the configuration and quantity of the HiNAS Control Standard hardware are the same as those shown in the drawings			v	
2	Verify that the Serial Numbers of the hardware listed in the DoC document match those of the hardware actually installed onboard.			v	

3.2 Functional Test

3.2.1 Dimmer

Verify the operation of the monitor's dimming function

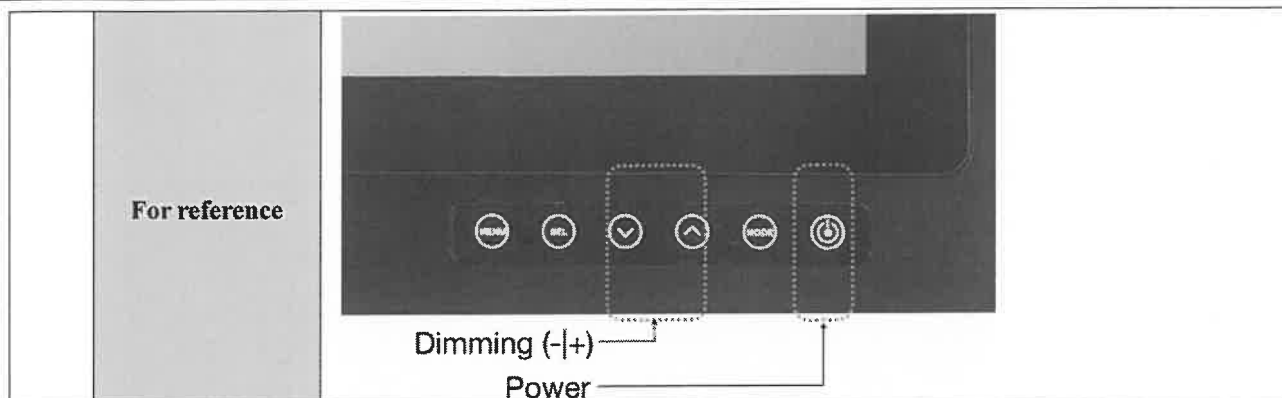
※ For monitors not applicable, indicate "N/A" in the Result field and omit the test

No.	Test	Criteria	Result	Judgment	
				Pass	Fail
1	Verify whether the brightness can be adjusted by pressing the “-” and “+” buttons on the OSD Remote Control Unit of model TIMX-UW584.	Brightness adjustable		v	
	For reference				

No.	Test	Criteria	Result	Judgment	
				Pass	Fail
2	Verify whether the brightness can be adjusted by pressing the “-” and “+” buttons located at the lower right corner of models TIMX-W270 and TIMX-W320	Brightness adjustable		v	



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3.2.2 Keyboard with Trackball

No.	Test	Criteria	Result	Judgment	
				Pass	Fail
1	Verify that the keyboard inputs function correctly.			v	
2	Verify that the mouse clicks function correctly.			v	

3.2.3 Software Installation Check Test

No.	Test	Criteria	Result	Judgment	
				Pass	Fail
1	Verify that HiNAS Control Standard SW is running on the monitor.	HiNAS Control Standard SW is running on the monitor.		v	
2	Double-click 'Avikus HiNAS' Icon on the desktop to open the login screen, log in with the provided default account	Login operates normally.		v	
3	Reboot the HiNAS Control Standard client PC and verify that the web browser automatically launches upon boot.	HiNAS Control Standard automatically launches after reboot.		v	
4	Verify the Ubuntu OS version of the server and client PCs using the 'lsb_release -a' command in the terminal.	Ubuntu 22.xx (x is 04 or higher)		v	
	For reference	<pre>avikus@avikus-10-0-100-20:~\$ lsb_release -a No LSB modules are available. Distributor ID: Ubuntu Description: Ubuntu 22.04.5 LTS Release: 22.04 Codename: jammy</pre>			



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5	Verify that the Version text at the lower left corner of HiNAS Control Standard matches the SW installation package.	Version v1.x.x (x is 0 or higher)		v	
	For reference	Environment Wind SPD T 0. Wind DIR T 83. CUR SPD T * CUR DIR T * Version v1.0.0 WGS 84 Mercator 19 20 # This is the version number of the application being deployed. This version number should be 21 # incremented each time you make changes to the application. Versions are not expected to 22 # follow Semantic Versioning. They should reflect the version the application is using. 23 # It is recommended to use it with quotes. 24 appVersion: "v1.0.0"			

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3.2.4 System Integration Test

All signals are applied to the firewall, network switch, and Serial to Ethernet device to verify the proper operation of the HiNAS Control Standard whole system. Software functions are being checked, and the related items can be verified in the 'Onboard Test Procedure document.'

1. Performance Tests
2. System Health Check Test
3. Basic Functionality Test
4. Route Planning Test
5. Autopilot fail test
6. Sensor interface fail test
7. BMS Fail Test
8. Power Fail Test
9. Operation Mode Change Test – Mode Selection Unit
10. Operation Mode Change Test – SW Button

3.2.5 Power failure test

Verify the operation of the UPS backup power when the input power is disconnected

No.	Test	Criteria	Result	Judgment	
				Pass	Fail
1	Upon disconnection of the input power to the UPS, the system shall remain operational via the UPS backup power.	Normal Operation upon UPS Input Power Disconnection		v	

3.2.6 Verification of Serial Number

No.	Test	Criteria	Result	Judgment	
				Pass	Fail
1	Verify the HINAS Control Standard serial number and the presence of the Wheel Mark by inspecting the nameplate on the server.	Serial Number and Wheel Mark Labeling		v	

RESULT		A	
“A”: Accepts, “B”: Accept with following comments, “C”: Canceled, “R”: Reinspection			
Signature	Avikus	DNV	
			



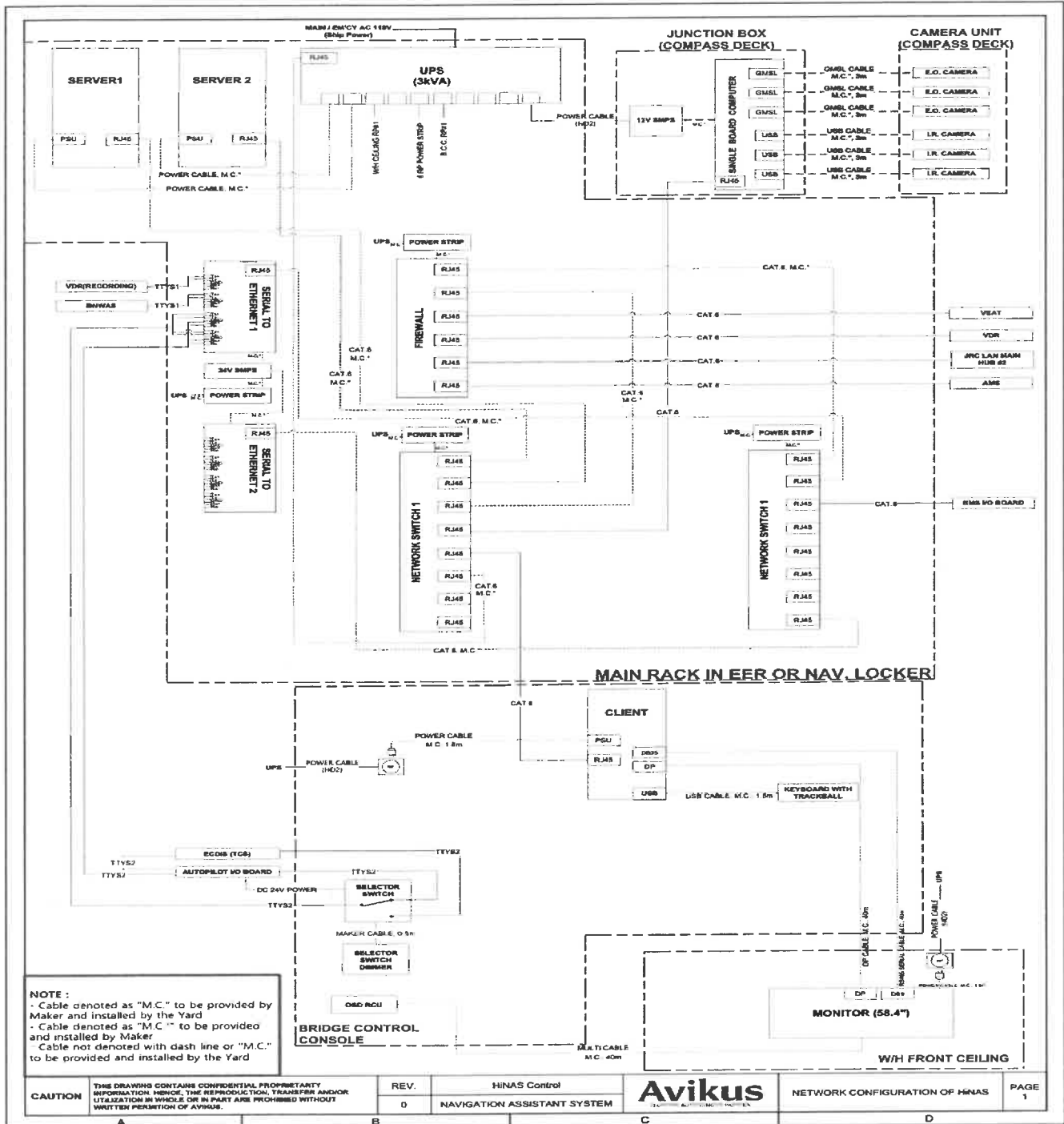
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Appendix A.

1. Wiring Diagram



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